

Eigenvalues and Eigenvectors

Math Foundations — Node 07

First, an example

Before any definitions, look at

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Read aloud: “A equals the upper-triangular two-by-two matrix with rows two one and zero three.” Its characteristic polynomial is

$$\det(A - \lambda I) = (2 - \lambda)(3 - \lambda) = 0,$$

Read aloud: “the determinant of A minus lambda I equals two-minus-lambda times three-minus-lambda equals zero.” so the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$. Solving $(A - 2I)v = 0$ yields the eigenvector $v_1 = (1, 0)^\top$, and $(A - 3I)v = 0$ yields $v_2 = (1, 1)^\top$. Each v_i is mapped by A to a pure rescaling: $Av_1 = 2v_1$ and $Av_2 = 3v_2$. The whole rest of this lesson is the systematic theory behind that little calculation.

1 Motivation

A linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is, in general, a complicated object: it mixes coordinates, rotates, shears, and rescales. The whole point of eigentheory is to look for the directions on which A acts trivially — not literally as the identity, but as pure scaling. Once such a direction v has been found, the action of A along it is described by a single number λ , the corresponding eigenvalue, via $Av = \lambda v$. If enough such directions can be found to form a basis, then A itself factorises as a change of basis followed by a diagonal scaling. This is the cleanest possible structural description of a linear map and it underlies essentially every spectral technique in numerical analysis, optimisation, and machine learning.

2 Definitions

Throughout, let $\mathbb{F}$ denote either $\mathbb{R}$ or $\mathbb{C}$ and let $A \in \mathbb{F}^{n \times n}$.

Definition 1 (Eigenvalue, eigenvector). A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of A if there exists a non-zero vector $v \in \mathbb{F}^n$ such that

$$Av = \lambda v.$$

Read aloud: “A times v equals lambda times v.” Any such v is called an *eigenvector* of A associated with the eigenvalue λ . The set of all eigenvectors for a fixed λ together with the zero vector is a linear subspace of $\mathbb{F}^n$ called the *eigenspace* $E_\lambda = \ker(A - \lambda I)$. The set of all eigenvalues of A is the *spectrum* $\sigma(A)$.

Remark. The condition $v \neq 0$ is essential. Without it, every scalar λ would satisfy $A \cdot 0 = \lambda \cdot 0$ trivially, and the definition would carry no information.

Definition 2 (Characteristic polynomial). The *characteristic polynomial* of A is

$$p_A(\lambda) = \det(A - \lambda I) \in \mathbb{F}[\lambda].$$

Read aloud: “p sub A of lambda equals the determinant of A minus lambda I, an element of F adjoin lambda.” It is a polynomial of degree exactly n in λ with leading term $(-1)^n \lambda^n$.

Definition 3 (Algebraic and geometric multiplicity). Let $\lambda_0 \in \sigma(A)$.

- The *algebraic multiplicity* $m_a(\lambda_0)$ is the multiplicity of λ_0 as a root of p_A .
- The *geometric multiplicity* $m_g(\lambda_0)$ is the dimension of the eigenspace $E_{\lambda_0} = \ker(A - \lambda_0 I)$.

It is always the case that $1 \leq m_g(\lambda_0) \leq m_a(\lambda_0)$.

Definition 4 (Diagonalisability). A is *diagonalisable* over $\mathbb{F}$ if there exist an invertible $V \in \mathbb{F}^{n \times n}$ and a diagonal $\Lambda \in \mathbb{F}^{n \times n}$ such that $A = V\Lambda V^{-1}$. Equivalently, $\mathbb{F}^n$ admits a basis of eigenvectors of A , equivalently $m_a(\lambda) = m_g(\lambda)$ for every $\lambda \in \sigma(A)$.

3 The fundamental theorem

We now state and prove the main structural fact tying eigenvalues to roots of the characteristic polynomial.

Theorem 5 (Eigenvalues are roots of the characteristic polynomial). *Let $A \in \mathbb{F}^{n \times n}$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A if and only if*

$$\det(A - \lambda I) = 0.$$

Read aloud: “the determinant of A minus lambda I equals zero.”

Proof. We prove both implications.

($\Rightarrow$). Suppose λ is an eigenvalue of A . By Definition 1 there exists a non-zero vector $v \in \mathbb{F}^n$ with $Av = \lambda v$. Rearranging,

$$(A - \lambda I)v = Av - \lambda v = 0.$$

Read aloud: “open paren A minus lambda I close paren times v equals A v minus lambda v equals zero.” Thus the homogeneous linear system $(A - \lambda I)x = 0$ has the non-trivial solution $x = v \neq 0$. A square matrix admits a non-trivial kernel vector if and only if it is singular, equivalently iff its determinant vanishes. Therefore $\det(A - \lambda I) = 0$.

($\Leftarrow$). Conversely, suppose $\det(A - \lambda I) = 0$. Then $A - \lambda I$ is a singular $n \times n$ matrix over $\mathbb{F}$, so its kernel $\ker(A - \lambda I)$ is a non-trivial subspace of $\mathbb{F}^n$ (it has positive dimension, by the rank-nullity theorem applied to the linear map $x \mapsto (A - \lambda I)x$, whose rank is at most $n - 1$). Choose any non-zero $v \in \ker(A - \lambda I)$. Then $(A - \lambda I)v = 0$, i.e. $Av = \lambda v$, with $v \neq 0$. By definition, λ is an eigenvalue of A with eigenvector v . $\square$

Corollary 6. *A matrix $A \in \mathbb{C}^{n \times n}$ has at least one eigenvalue in $\mathbb{C}$. In fact it has exactly n eigenvalues counted with algebraic multiplicity, because $p_A \in \mathbb{C}[\lambda]$ splits completely over $\mathbb{C}$ by the fundamental theorem of algebra.*

4 The spectral theorem (symmetric case)

Theorem 7 (Real spectral theorem). *If $A \in \mathbb{R}^{n \times n}$ is symmetric ($A^\top = A$), then all eigenvalues of A are real, and there exists an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ (i.e. $Q^\top Q = I$) and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ such that*

$$A = Q\Lambda Q^\top.$$

Read aloud: “ A equals Q big-lambda Q transpose.”

We do not prove this here but record it because it is the workhorse fact for covariance matrices, Gram matrices, and Laplacians.

5 A small worked example

Example 8. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Read aloud: “ A equals the two-by-two matrix with rows two one and one two.” Then

$$p_A(\lambda) = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Read aloud: “ p sub A of λ equals the determinant of the matrix with rows two-minus-lambda one and one two-minus-lambda, which equals two-minus-lambda squared minus one, equals lambda squared minus four lambda plus three, equals lambda-minus-one times lambda-minus-three.” Thus $\sigma(A) = \{1, 3\}$. Solving $(A - I)v = 0$ gives $v_1 = (1, -1)^\top / \sqrt{2}$, while $(A - 3I)v = 0$ gives $v_2 = (1, 1)^\top / \sqrt{2}$. These are orthonormal, as guaranteed by Theorem 7, and

$$A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Read aloud: “ A equals one-half times the matrix with rows one one and minus-one one, times the diagonal matrix one-three, times the matrix with rows one minus-one and one one.”

6 Where to go next

The next step in this map is the singular value decomposition (SVD), which extends eigendecomposition to rectangular and non-normal matrices and underlies LoRA-style low-rank adaptation, PCA, and dimension reduction in general.